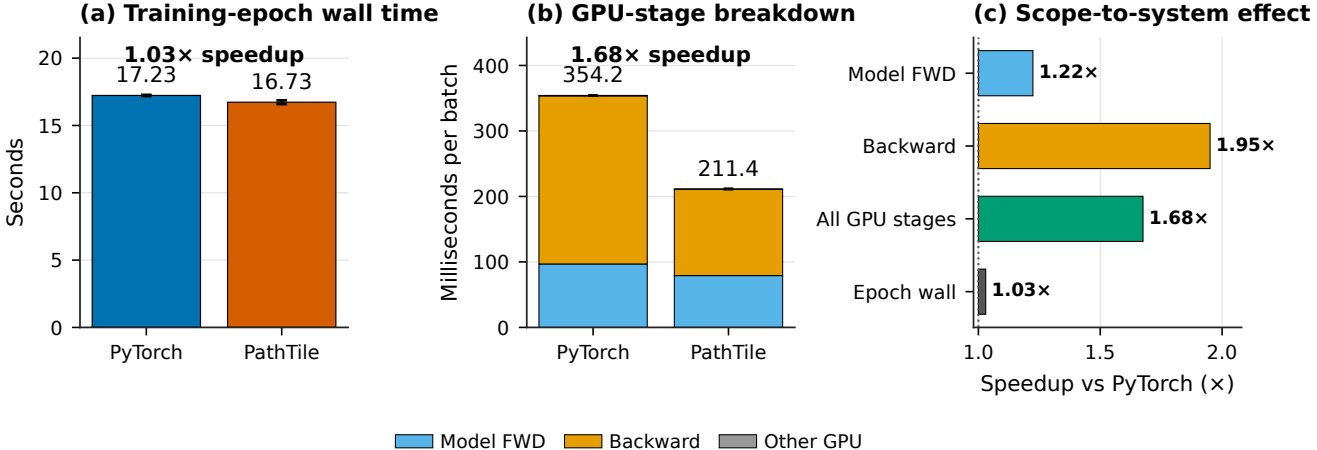
(a) Training-epoch wall time
1.03× speedup
17.23
16.73
Seconds
0
5
10
15
20
PyTorch
PathTile
(b) GPU-stage breakdown
1.68× speedup
354.2
211.4
Milliseconds per batch
0
100
200
300
400
PyTorch
PathTile
(c) Scope-to-system effect
Model FWD
Backward
All GPU stages
Epoch wall
1.22×
1.95×
1.68×
1.03×
1.0
1.5
2.0
Speedup vs PyTorch (×)
Model FWD
Backward
Other GPU